SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Ganji Madhan kumar	**Reg. No.:**	192412328
Section / Batch:		**Date:**	11-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:		**100 Marks**

SECTION A – CONCEPT MAPPING

Code of Conduct

I ____________________________ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________________________

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	**Concept Identification (25%)**	**Linkages (25%)**	**Organization (20%)**	**Integration of Literature (20%)**	**Clarity (10%)**	**Total Marks (Each – 50)**
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge

Signature: ____________________

Date: ____________________

Course Coordinator

Signature: ____________________

Date: ____________________

Head of Department

Signature: ____________________

Date: ____________________

CO1_AT02

- Concept Mapping

Name: G.Madhan kumar
Reg No: 192472328
CODE: CSA0624
Course: DAA

ALGORITHM

A step-by-step procedure to solve a problem

TIME COMPLEXITY

Measurement of time taken by an algorithm as a function of input size(n)

BEST_CASE ANALYSIS

Minimum time taken for the most favorable input of size n

AVERAGE_CASE ANALYSIS

Average time taken over all possible inputs of size n

WORST_CASE ANALYSIS

Maximum time taken for the least favorable input of size n

ASYMPTOTIC NOTATION

Mathematical way to describe the growth rate of time complexity as $n \to \infty$

BIG-O(O)

Upper Bound

$T(n) = O(f(n))$

Represents the worst-case growth rate (asymptotic upper limit)

BIG-THETA(Θ)

Tight Bound

$T(n) = \Theta(f(n))$

Represents the exact growth rate

BIG-OMEGA(Ω)

Lower Bound

$T(n) = \Omega(f(n))$

Represent the best-case growth rate.

Algorithm → Time Complexity: An algorithm's abstract steps get counted as a function of input size n.

Time Complexity → Best / Average / Worst Case: Splits by which input - best-case (fewest ops), average-case (typical ops), worst-case (Max ops).

→ Asmptotic Notation: Strips away constants / hardware noise, keeps. only growth trend.

→ Big-O / Big-θ / Big-Ω: Expresses that growth precisely - O = Upper bound, Ω = lower bound, Θ = tight bound (both match).